METHOD FOR THE HYDROCLIMATOLOGICAL CHARACTERIZATION AND HYDROLOGICAL ASSESSMENT OF THE NATURAL WATERSHEDS IN THE MINING AREA OF CENTRAL CESAR STATE, COLOMBIA. MEAN, MAXIMUM AND MINIMUM FLOWS

ALFONSO RODRÍGUEZ DÍAZ(1), RICARDO AGUILAR PIÑA(2) , ACERO RIVEROS GERMÁN(3), GERMÁN SANTOS GRANADOS(4) & ROMEO RAMOS(5)

(1) *Escuela Colombiana de Ingeniería Julio Garavito, Bogotá, Colombia, alfonso.rodriguez@escuelaing.edu.co*
(2) *Escuela Colombiana de Ingeniería Julio Garavito, Bogotá, Colombia, william.aguilar@escuelaing.edu.co*
(3) *Escuela Colombiana de Ingeniería Julio Garavito, Bogotá, Colombia, german.acero@escuelaing.edu.co*
(4) *Escuela Colombiana de Ingeniería Julio Garavito, Bogotá, Colombia, german.santos@escuelaing.edu.co*
(5) *Drummond Ltd., Bogotá, Colombia, RRamos@drummondltd.com*

ABSTRACT

A methodology was developed to evaluate the surface water availability based on the requirements established by Drummond Ltd, which is in charge of the surface mining development in the south of the State of Cesar, Colombia. The methodology includes a long term Water Balance to estimate the mean flow in any point of the watershed drainage network and the determination of the maximum flow, for a given return period, in previously defined points.

The technique developed to estimate the different kinds of flows is supported by a GIS and commercial as well as public domain software. In addition, new tools were developed to evaluate hydrological processes that can be used in any other watershed system.

The first research phase deals with the surface water availability for the natural conditions. The region under study has an area of more than 4000 km2 and 13 mayor watersheds were defined. In the next phases, several particular studies were made to evaluate the water availability according to the transformations occurred.

The GIS implemented and structured has a spatial model that covers the entire drainage network in natural conditions a series of hydro climatological variables in a spatial extend to the watershed divide in the Sierra del Perijá in The East Andes.

The model includes, based on 24 hours maximum rainfall depth, a methodology to generate rainfall and construct hyetographs. It is possible to classify the soils according to the US Soil Conservation Service methodology to obtain the required Curve Numbers in the rainfall- runoff transform process.

To estimate the mean flow, a long term water budget balance was performed using the ArcGis- Esri spatial analysis tool in each cell for the entire study zone based on the mean precipitation and evaporation grids.

The developed methodology allows to perform analysis and to obtain results for the extreme flows and hydrographs at any point of a channel in the mining development zone. The maximum flows are estimated by the HEC-HMS model based on the precipitation data and the physical characteristics previously obtained and available in the GIS and processed by HEC-GeoHMS. The channel routing is made hydrological by the Muskingum method.

In conclusion, a model was developed that reproduce the natural conditions and to support decisions about the water management in the mining development.

Keywords: Hydro climatological characterization, surface water availability, mean flow estimation, long term water budget balance, extreme flows.

1. INTRODUCTION

The Escuela Colombiana de Ingeniería, Julio Garavito, through its Center of Hydraulic Studies, has developed a methodology to assess the surface hydric offer based on the request made by Drummond Ltd. and the requirements established by the same Company, responsible of open-pit mining developments around hydrographic basins with relatively few hydrological information that make up the influence zone of the Southern mining area of Cesar (Colombian department). The methodology contains a long-term hydric balance to estimate mean flows at any point of a basin's drainage grid, and maximum flow calculation for a determined return period, at previously defined points.

The goal of this Project is to evaluate the surface hydric offer in the basins that make up the influence area of the mining zone developed by Drummond Ltd. (Long-term hydric balance for mean discharges. Minimum flows and maximum flows) for natural conditions, through conventional hydrology methodologies, agreed with Drummond Ltd. and according to available information.

©2015, IAHR. Used with permission / ISBN 978-90-824846-0-1

Henceforth, a detailed general scheme of the developed work with this assessment methodology of surface runoff.

2. INFORMATION COLLECTION

Based on available information, hydroclimatic variables data from the Cesar mining area where Drummond Ltd. works were collected, until the divide of the Eastern mountain range in the serranía del Perijá, with the objective of structuring and assembling a spatial model for all of the hydric network, through a Personal Geodatabase (on the ESRI ARCSIG platform). Besides, the company owns a network of hydroclimatic stations that were incorporated as points in the base of the current work.

To establish the soil coverage, information used to determine the runoff curve numbers (CN), references were taken from zoning soil studies of Colombia and the department of Cesar made by Instituto Geográfico Agustin Codazzi (IGAC). Orthoimages from the study zone provided by the Global Land Cover Facility service of the University of Maryland were used, as well as the digital elevation model obtained from NASA. Parameters and data collected from different sources were integrated in a relational database, so that the data model can include every discreet data in a register with identification pointers by link levels.

The natural drainage network, that is, before mining operations, is composed by 570 streamflow reaches with an approximate length of 2,543 kilometers.

3. ANALYSIS OF AVAILABLE HYDROLOGICAL AND HYDRAULIC STUDIES. PERFORMED WORK ON GIS.

One of the objectives of this research work is establishing and developing a hydrologic or hydraulic model based on a Geographical Information System that allows to calculate surface runoff in mean and extreme flows. Therefore, field information and available studies make up a primary source of contrast for short, mid, and long term calibration of the model.

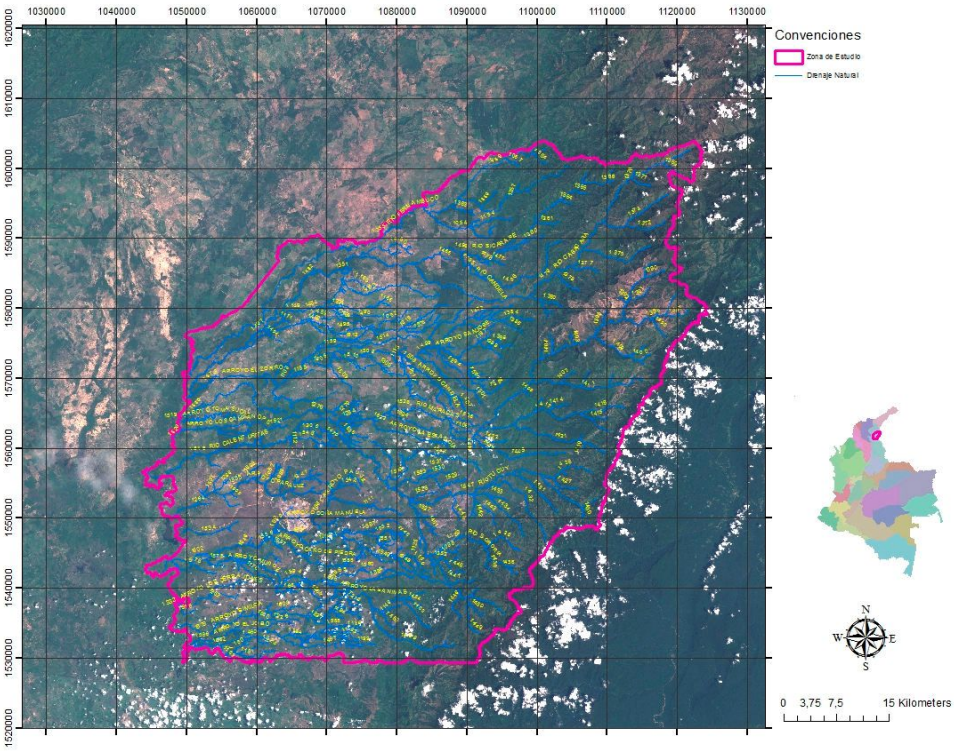
In the study zone, Drummond Ltd. has been building a hydrological information network that constitutes the support for the quality of the hydrologic data obtained from the developed numerical model.

The hydrologic and hydraulic information is mostly composed by prevoius works for environmental impact studies in the project zone, of mean flows, and those that deal with realigning some natural streams and complementary works in different basins.

3.1 Implementation of the digital elevation model for the study area

To delimit the basins based on the digital elevation model, a grid that covered all of the study zone was used. The morphometric characterization of basins was made with the tools available in ArcGIS 10.

Similarly, the natural drainage network was digitized from the photointerpretation of orthophotographs and Landsat satellite images. Information was edited establishing the streams between affluent junctions and tagging creeks, streams, and main major rivers, to obtain a base drainage system (figure 1).



©2015, IAHR. Used with permission / ISBN 978-90-824846-0-1

Figure 1. Base drainage system for the natural-state conditions.

Based on the digital NASA model, the ground was reconditioned in order to embed the natural drainage network photo-restored to specify the flow direction and accumulation, to help delimit the sub-basins. Contribution area was specified as the equivalent value of four square kilometers in cell sum.

Taking into account the assessed drainage system, it was possible to determine thirteen main basins, used as a reference for the analyses in the rainfall runoff transformation. The main studied basins are shown below (figure 2).

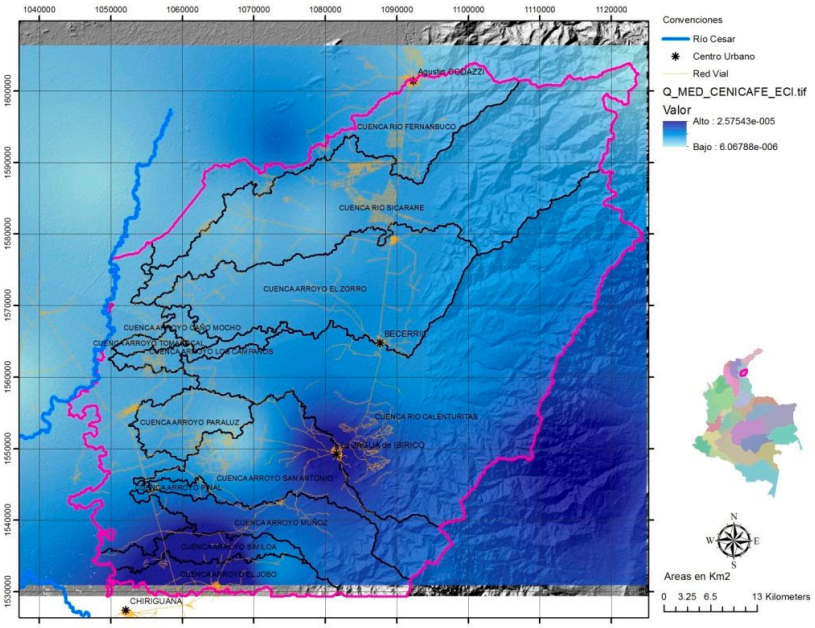


Figure 2. Main basins determined for the hydrological study.

3.2 Analyses of precipitation and hydro climatological variables

3.2.1 Mean multiannual total precipitation

Precipitation logs of 20 stations located in and around the interest zone were used as subsequently indicated (figure 3).

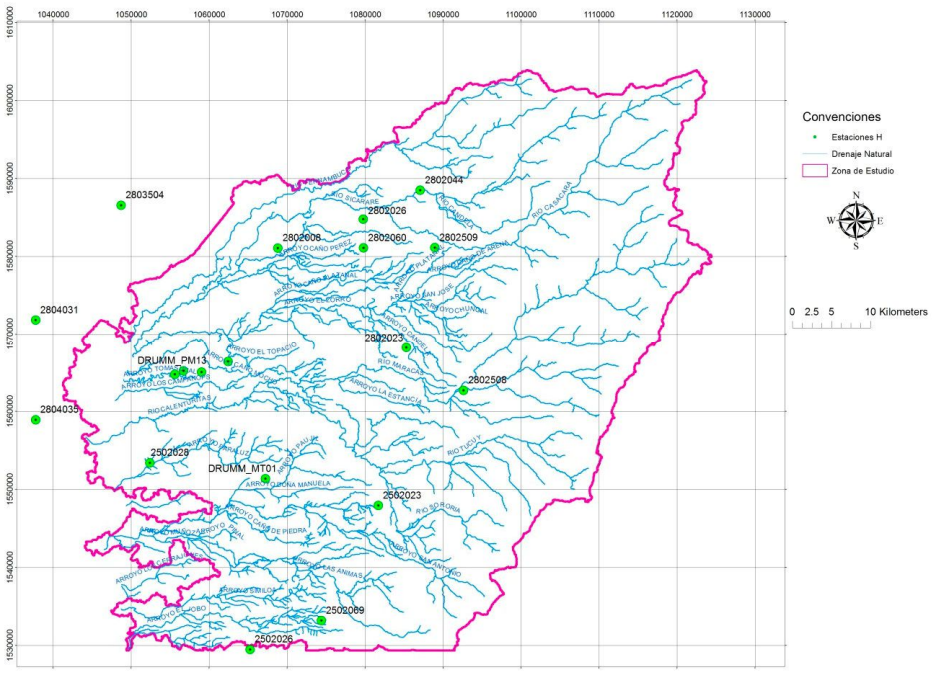


Figure 3. Precipitation stations used in the study.

Due to limitations in spatial information and with the purpose of obtaining maps for all of the studied area, it was necessary to define virtual stations with known mean multiannual values, located in the extreme of the studied area.

corresponding to high mountain zones. To define the data of these virtual stations the study of mean precipitations for Colombia was used.

In a similar way, mean annual values were estimated for the 1981-2003 period, which corresponds to a 23-year series. From 2003 on, the anthropic condition was defined considering that streams intervention, as a byproduct of mining activities, started from that moment. The results of the analysis made for mean precipitation will now be presented, in the project area in natural conditions (figure 4).

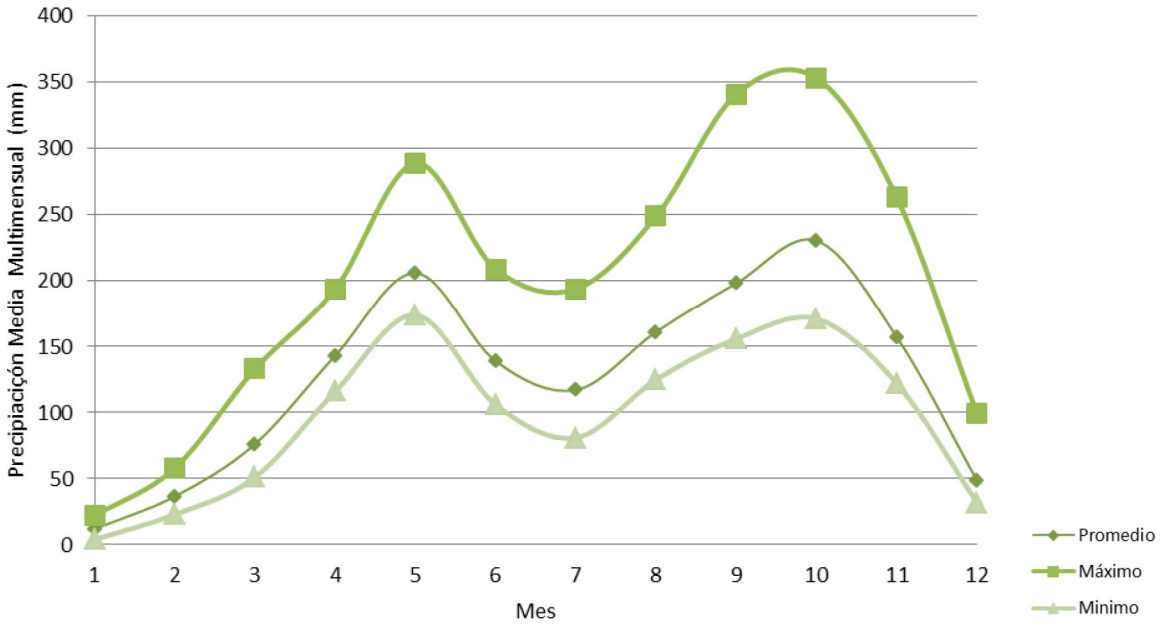


Figure 4. Mean, maximum, and minimum multi-monthly precipitations [mm] in study zone from total monthly precipitation, 1981-2003 series.

Finally, based on the obtained precipitation results and the methodologies employed by SIG for interpolation and obtaining mean precipitation grids, this information was chosen to calculate the hydric long-term balance (annual) (figure 5).

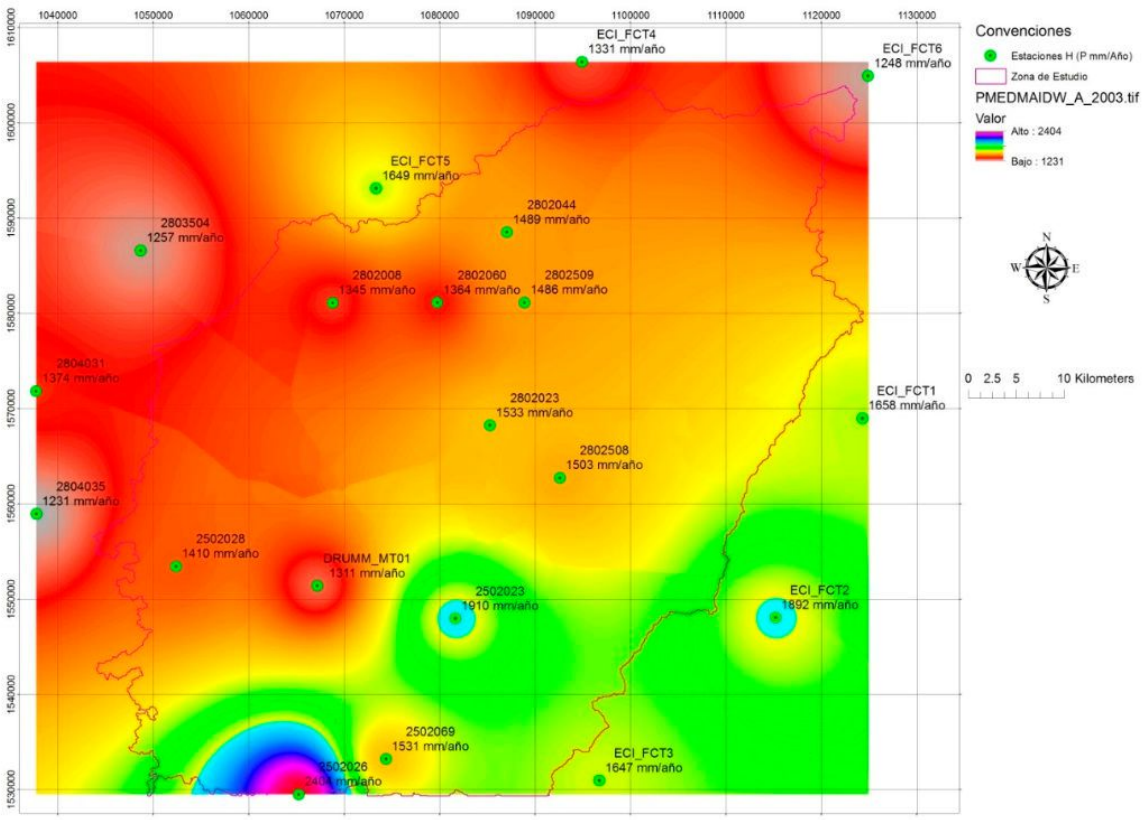


Figura 5. Grid for mean multiannual values [mm], ECI IDW. 1981-2003 series.

©2015, IAHR. Used with permission / ISBN 978-90-824846-0-1

3.2.2 Maximum precipitation in 24 hours

Analogous to the total mean multiannual precipitation analysis, data series from the 1981-2003 period were used, and based on the results, a map of mean multiannual maximum precipitation values in 24 hours was generated.

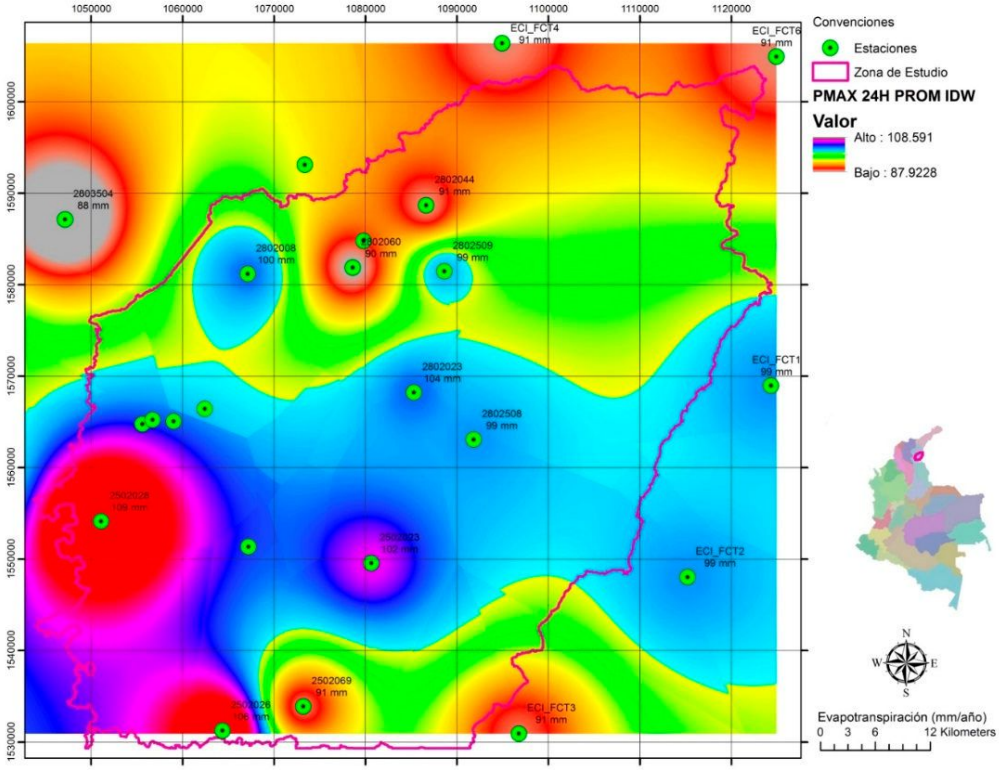


Figure 6. Grid of mean multiannual maximum precipitation values in 24 hours [mm].

Founded on the resulting maps, grids for maximum precipitation in 24 hours were made for different return periods.

3.2.3 Intense rains. Temporal distribution

The temporal distribution of rain analysis with a return period, from the information available at the centroid of a basin's area, was done based on a previous storms study from the zone. As a result of the analysis, the behavior of intense rains in the basin, a trend line (dotted), and the corresponding equation are shown (figure 7).

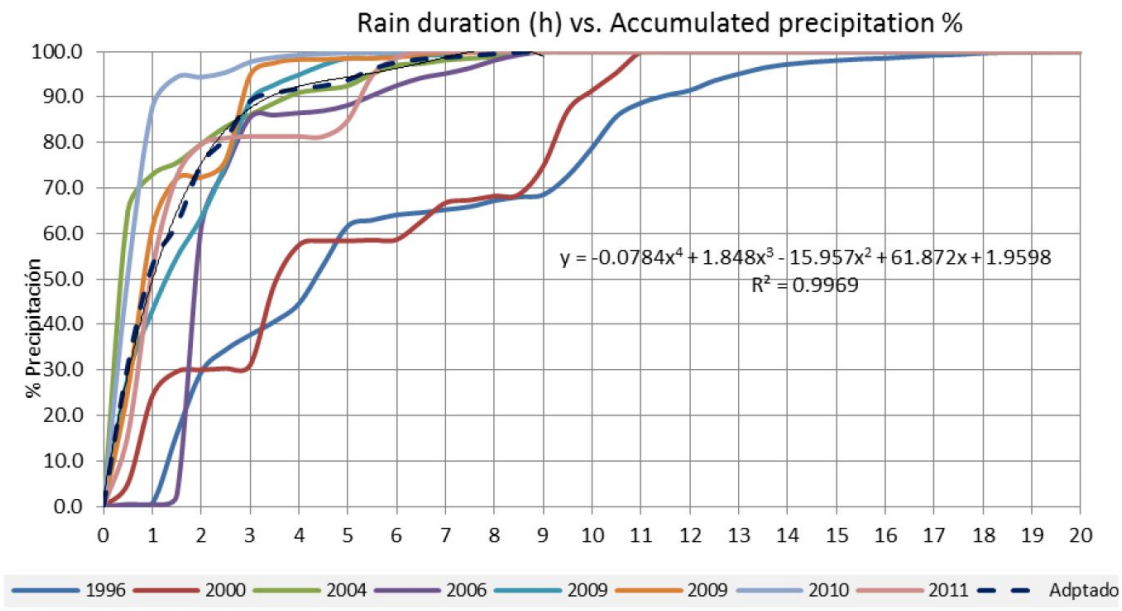


Figure 7. Temporal behavior of intense rains.

3.2.4 Intense rains. Areal or drainage size attenuation

In the process of hydrological simulation, basins with areas ranging from 4 to 1,000 km2, and even bigger, were taken into account, which is why the design rain is not presented uniformly and simultaneously in all of the drainage area. Therefore, calculation procedures are required to calculate an attenuation coefficient as a function of the basin's size.

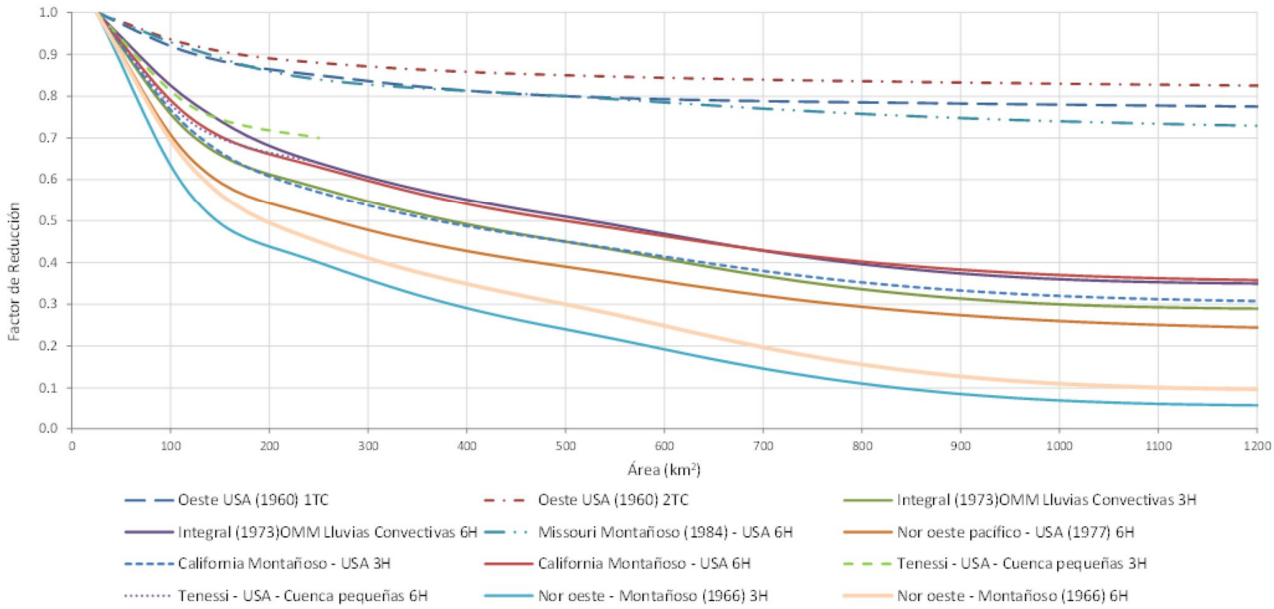
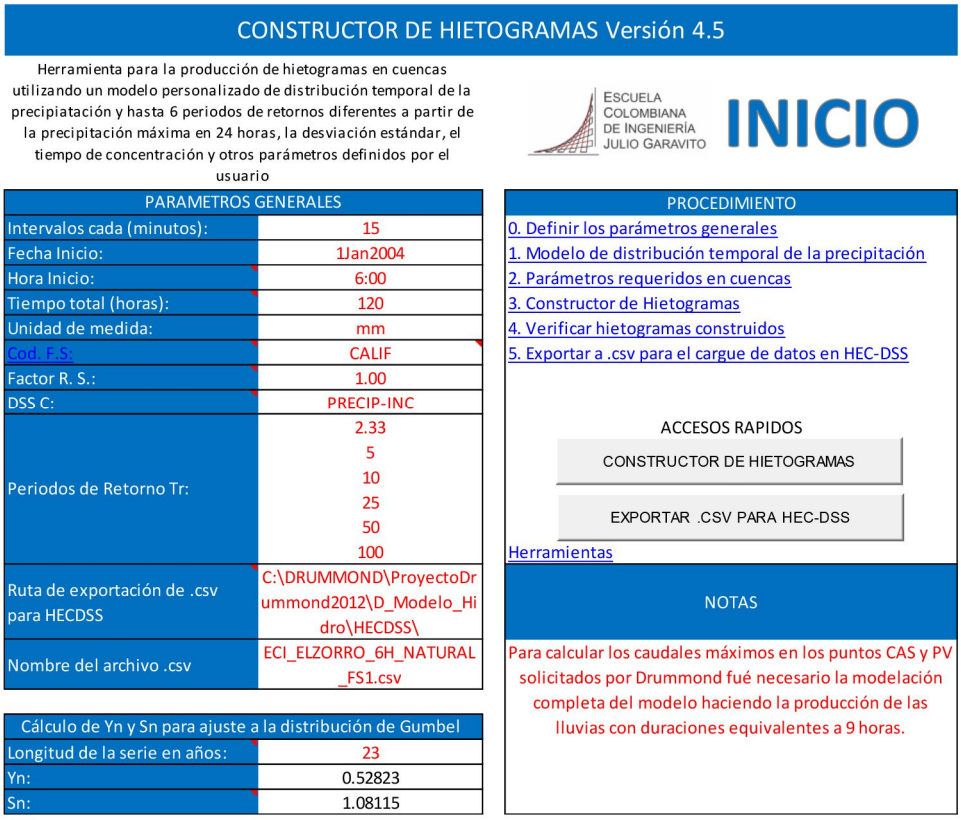


Figure 8. Precipitation reduction factor as a function of basin area.

3.2.5 Generation of design hyetographs

Since the available information regarding intense rains is very scarce, a program to generate design hyetographs was developed, based on the attenuation coefficients calculated according to established criteria previously presented (figure 9).



CONSTRUCTOR DE HIETOGRAMAS Versión 4.5

Herramienta para la producción de hietogramas en cuencas utilizando un modelo personalizado de distribución temporal de la precipiatación y hasta 6 periodos de retornos diferentes a partir de la precipitación máxima en 24 horas, la desviación estándar, el tiempo de concentración y otros parámetros definidos por el usuario

ESCUELA COLOMBIANA DE INGENIERÍA JULIO GARAVITO

INICIO

PARAMETROS GENERALES	
Intervalos cada (minutos):	15
Fecha Inicio:	1Jan2004
Hora Inicio:	6:00
Tiempo total (horas):	120
Unidad de medida:	mm
Cod. F.S:	CALIF
Factor R. S.:	1.00
DSS C:	PRECIP-INC
Periodos de Retorno Tr:	2.33 5 10 25 50 100
Ruta de exportación de .csv para HECDSS	C:\DRUMMOND\ProyectoDrummond2012\D_Modelo_Hidro\HECDSS\
Nombre del archivo .csv	ECI_ELZORRO_6H_NATURAL_FS1.csv

Cálculo de Yn y Sn para ajuste a la distribución de Gumbel	
Longitud de la serie en años:	23
Yn:	0.52823
Sn:	1.08115

PROCEDIMIENTO

0. Definir los parámetros generales
1. Modelo de distribución temporal de la precipitación
2. Parámetros requeridos en cuencas
3. Constructor de Hietogramas
4. Verificar hietogramas construidos
5. Exportar a .csv para el cargue de datos en HEC-DSS

ACCESOS RAPIDOS

CONSTRUCTOR DE HIETOGRAMAS

EXPORTAR .CSV PARA HEC-DSS

Herramientas

NOTAS

Para calcular los caudales máximos en los puntos CAS y PV solicitados por Drummond fué necesario la modelación completa del modelo haciendo la producción de las lluvias con duraciones equivalentes a 9 horas.

Figure 9. Hyetograph generator for the developed model.

©2015, IAHR. Used with permission / ISBN 978-90-824846-0-1

3.2.6 Temperature analysis

Besides the temperature information from the stations in plain lands within the project's influence area, the one compiled by the hydro climatological studies of Cenicafé were also used to generate a temperature map, whose information is useful for determining evapotranspiration.

3.2.7 Evapotranspiration analysis

To make the long-term hydric balances and estimate mean annual flows for the basins, real and potential evapotranspiration maps were generated. To calculate potential evapotranspiration, the formula from Cenicafé's climatological studies was used, and to calculate real evapotranspiration, Turc equation was used. Following up, the map corresponding to real evapotranspiration is shown (figure 10).

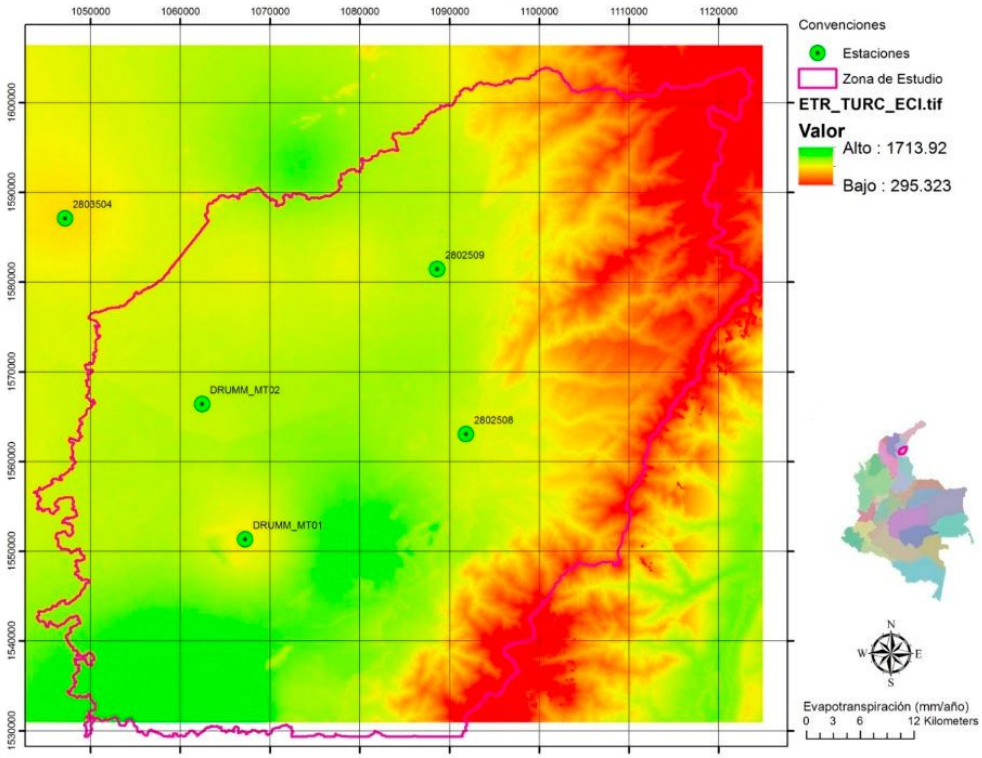


Figure 10. Real evapotranspiration grid [mm] from Cenicafé's EVP and Turc equation.

3.2.8 Soil classification using the methodology proposed by the U.S. Soil Conservation Service

The methodology proposed by the Soil Conservation Service (SCS), from the Department of Agriculture of the United States, allows to estimate the losses by infiltration and ponding, or initial loss, in the rainfall runoff analyses, as well as net precipitation. Such calculation process is done through assigning a factor known as CN (curve number) that is determined from several factors, such as: soil cover –in terms of permeability-, land use, the hydrological conditions –related to the coverage grade-, and the antecedent moisture condition.

Since it is a vast zone of over 7,000 km2, regional information was used, produced mostly by IGAC. Cartography of the interest area referent to the agrological soil types was employed, as well as maps of land use to create a grid or map of CNs employing HEC_GEO_HMS (figure 11).

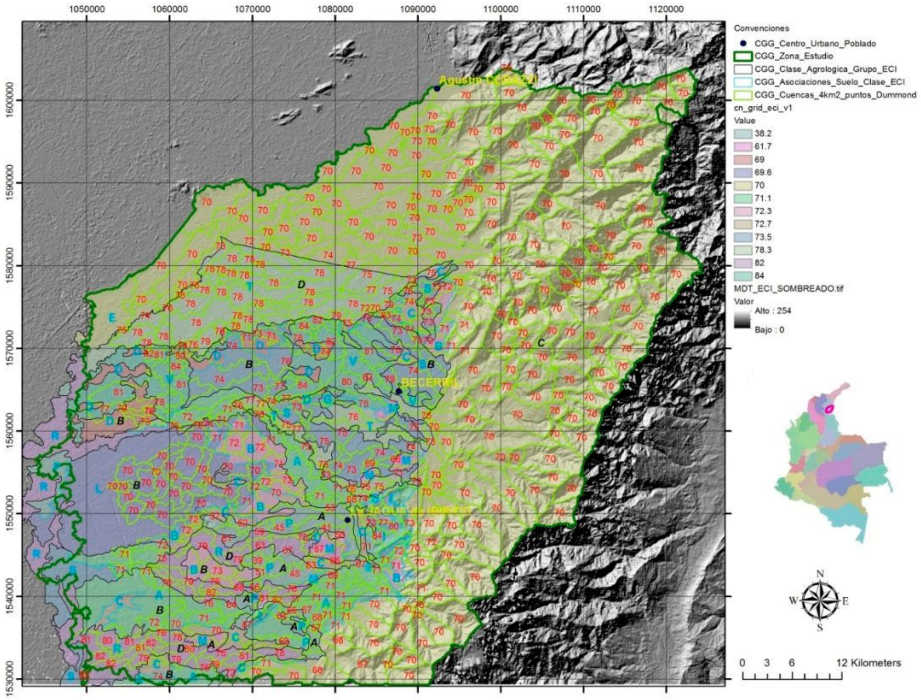


Figure 11. Mean CN values by sub-basin

4. MEAN FLOW ANALYSIS

To estimate mean flows, a long-term hydrological balance in each cell within the study area was done based on the maps of mean precipitation and evapotranspiration, which allows to determine mean flows at any point of the drainage network on the map. The map for mean flows generated by the model are now shown, some interest points for Drummond Ltd. are pointed out (figure 12).

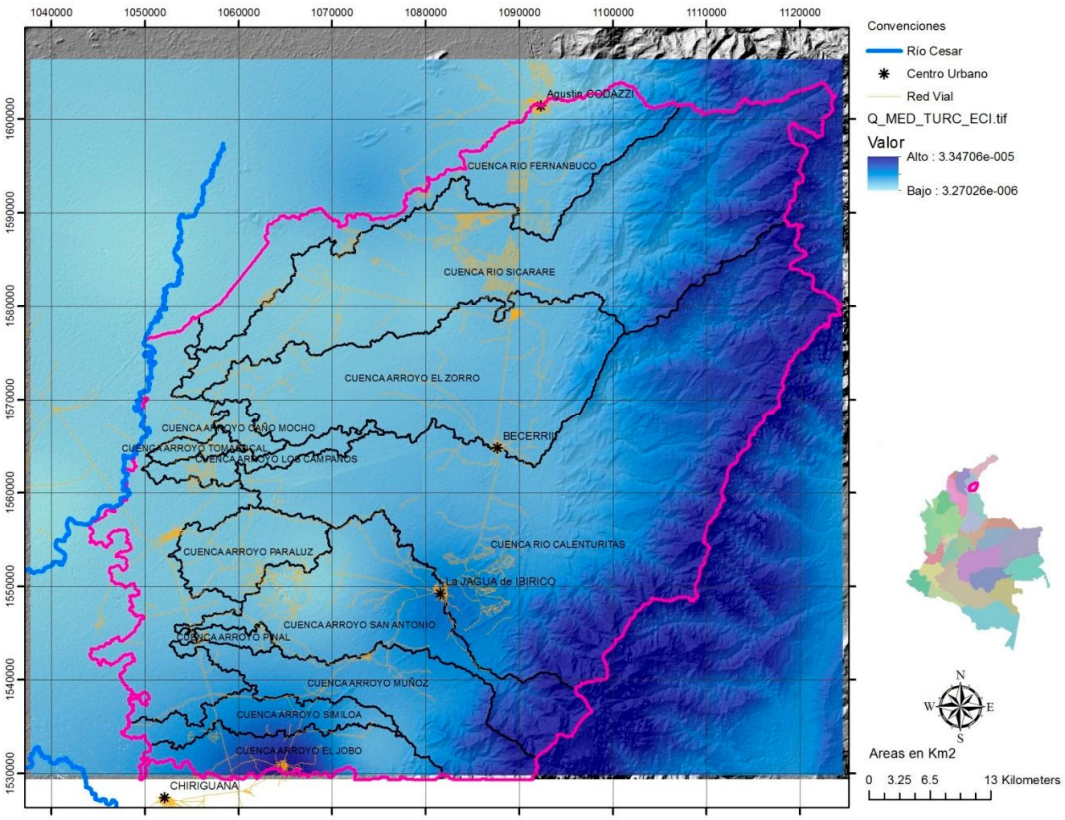


Figure 12. Study points and grid of mean flows.

©2015, IAHR. Used with permission / ISBN 978-90-824846-0-1

Based on the reading of mean flows on the 29 Drummond points of study, the results were also assessed in all 229 additional points of drainage confluence in the study zone, in order to build an interpolated map of mean flows iso-yield (figure 13).

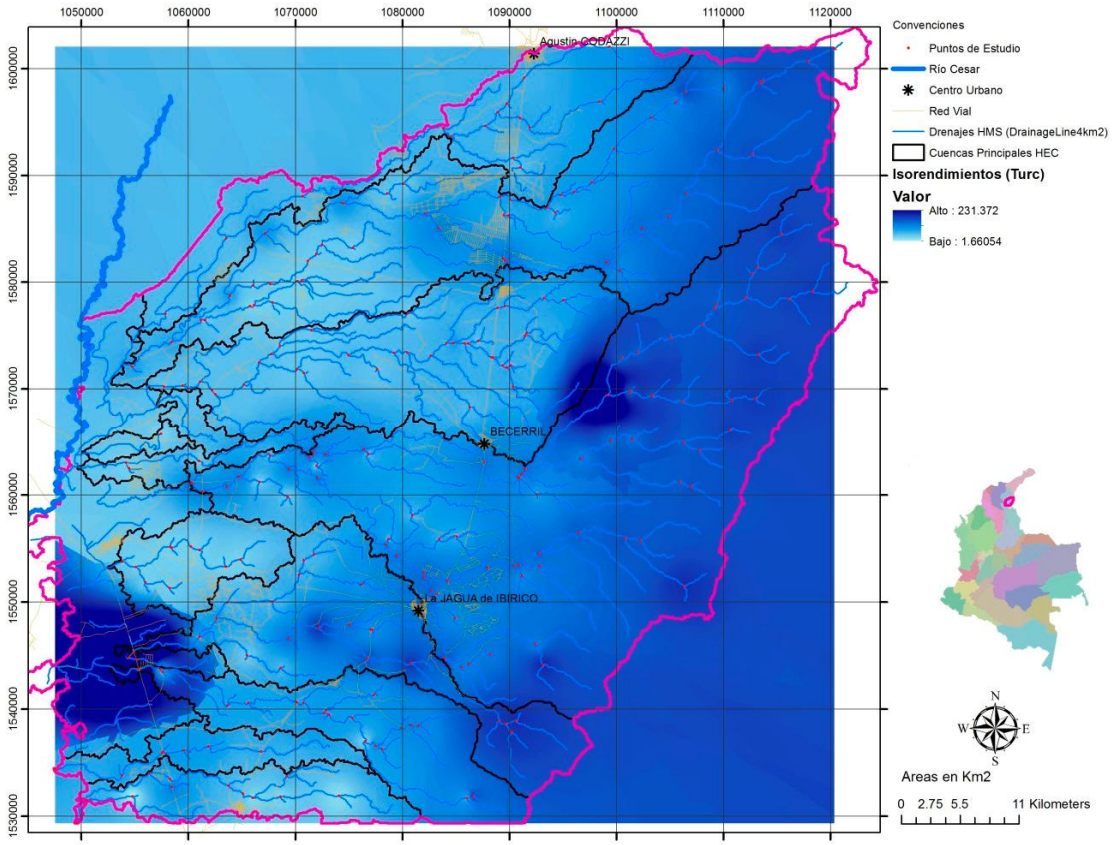


Figura 13. Iso-yield grid.

5. EXTREME FLOWS

Since the developed model is supported on a SIG, it is possible to determine extreme flows at any point of the drainage network. To achieve this it is necessary, as a previous work, to establish the drainage area and its corresponding drainage grid that allows the construction of the hydrological model in HEC-GEO-HMS, used in the HEC-HMS model. Later in this paper one of the generated models is shown (figure 14).

Especially for each one of the main basins determined in this study, different physical and hydrological parameters were calculated for each one of its sub-basins, which are necessary in the execution of the developed model. In a similar manner, for each one of the river streams considered in every main basin, the parameters used in the Muskingum method for channel routing were estimated.

For this particular project, the methods selected for modelling were those of SCS for rain losses and for rainfall runoff transformation and Muskingum for flood routing.

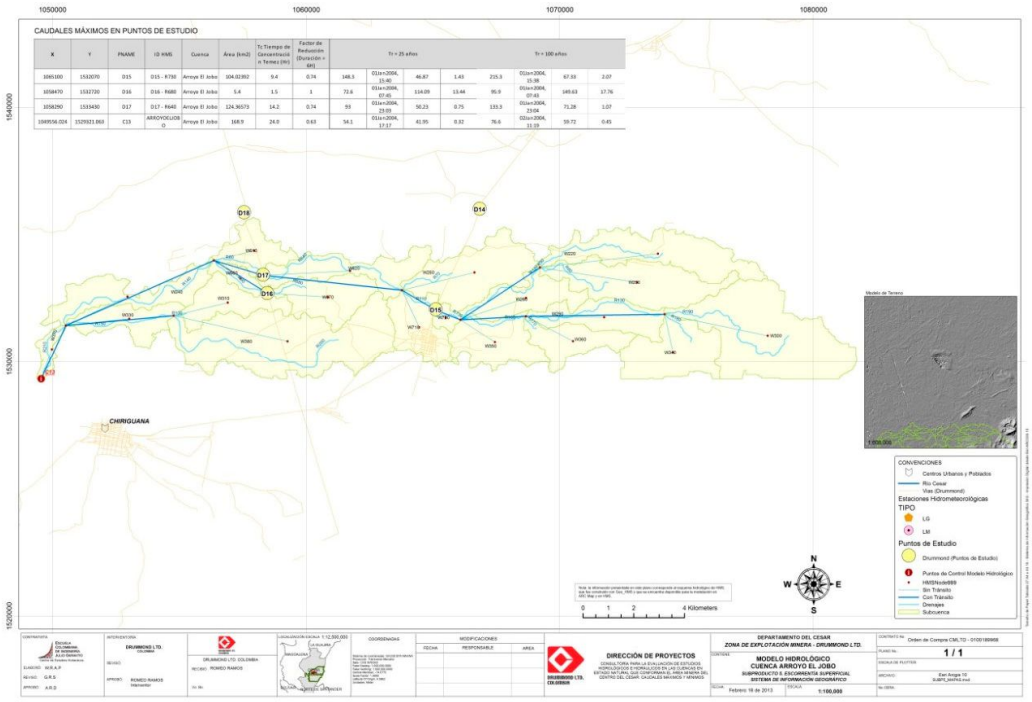


Figure 14. Hydrological model scheme generated with HEC-GEO-HMS for one of the main basins.

In rainfall runoff transformation using the HEC-HMS model, the effective precipitation must be determined, to achieve this, the US SCS curve number method, which establishes effective precipitation based on the CN was employed. Hydrological routing was done with the Muskingum method, which uses the continuity equation and ignores lateral flow. As mentioned, initial modelling in HEC-HMS was done for all thirteen basins in the study.

Calibration of the model's results was done with the basin of El Zorro stream that has field information whose results allow contrasting with the obtained results in this HMS model.

A hydrograph is now presented with the model's results (figure 15).

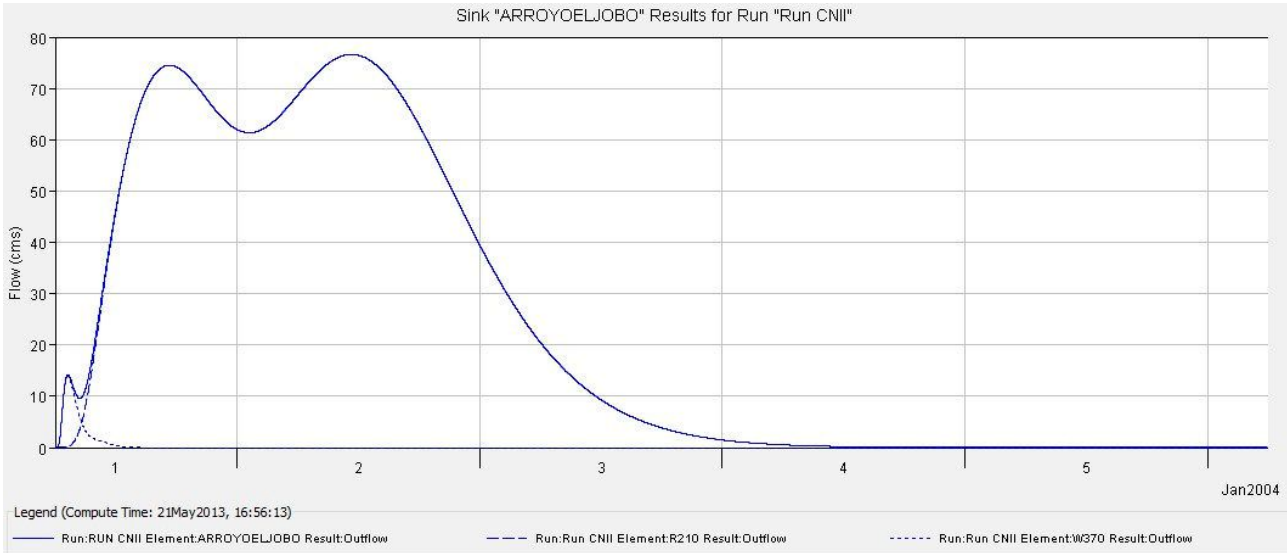


Figure 15. Rain-surface runoff transformation, model's results. Hydrograph.

Minimum flows were analyzed based on information from the three limnimetric stations available for the project area: Becerril Station, on the Maracas River, Islandia Station on Calenturitas River, and Santa Teresa Station on the Sicarare River.

Although the purpose was to make a spatial analysis, the lack of available information only allowed to make some punctual estimations for the stations' sites and perform certain qualitative considerations for all the zone.

 ©2015, IAHR. Used with permission / ISBN 978-90-824846-0-1

6. CONCLUSIONS

1. A rain-surface runoff model was built, supported on a geographical information system which shows that, once integrated to the information, it is possible to easily determine multiple scenarios with multiple models.
2. A model for hyetographs generation was developed, despite scarce hydrological information. This model takes into account the spatial and temporal distribution of the rain.
3. The developed model is a dynamic one, it fundamentally depends on information proceeding from hydrological data being input.
4. HEC-GEO-HMS module from ArcGIS is a tool that facilitates the rain-surface runoff transformation through the use of all available geographical information.
5. For the concrete case of hydrographical basins that make up the Southern mining zone of Cesar (Colombia) valuable information from mean and maximum flows is available at any point of the system.

ACKNOWLEDGMENTS

The Centro de Estudios Hidráulicos of Escuela Colombiana de Ingeniería wants to special thanks to Drummond Ltd. for its unconditional support and sponsoring in this project. The information they provided and their recommendations were very important in achieving the goals of this work.

REFERENCES

Chow, V.T. et al (1988). Applied Hydrology. Ed. Mc Graw Hill Book Company, NY, 201-324.
Drummond Ltd. (2004-2014). Collected information corresponding to more than 30 consulting works about watersheds hydro climatological characterization. Watersheds in the mining area of Central Cesar State, Colombia.
Esri (2010). ArcGIS 10.
Landsat (2001). Orthorectified Landsat digital data.
NASA. Digital elevation models, WGS 1984.
US. Army Corps of Engineers. Hec HMS 4.0.